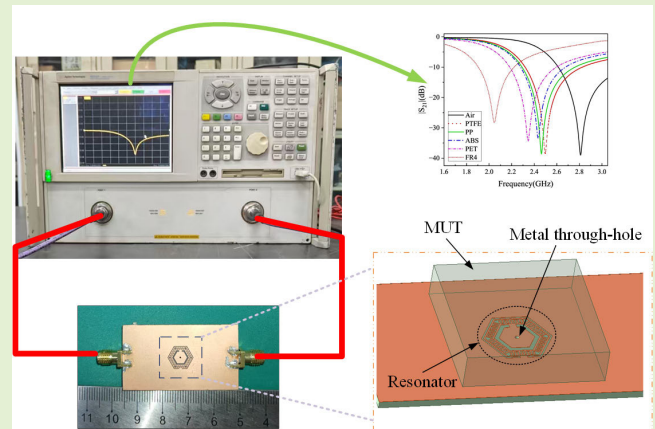


A Novel Microwave Sensor With Enhanced Coupling to Measure Dielectric Constants of Solid Materials

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Abstract—A complementary split-ring resonator (CSRR) planar microwave sensor loaded with interdigital electrode (IDE) is proposed to measure complex dielectric constants of solid materials. The sensor consists of a CSRR-IDE structure etched on the bottom layer of a 1-mm F4B substrate with a metal through-hole between a microstrip line and the resonant structure. The IDE structure enhances electric fields to achieve high-sensitivity measurement. The metal through-hole enhances the coupling and weakens the relationship between $|S_{21}|$ and the real part of the complex dielectric constant, improving the accuracy of the imaginary part measurement. The experimental results show that the proposed sensor has a high sensitivity of 9.6%, and the measurement errors of the real and imaginary parts of the complex dielectric constant are 1.3% and 3.0%, respectively. The microwave sensor proposed in this article achieves better results than most conventional sensors with similar structures.

Index Terms—Complementary split-ring resonators (CSRRs), interdigital electrode (IDE), microwave sensor.



I. INTRODUCTION

WITH the rapid advancement of electronic technology, microwave systems have been widely adopted in diverse fields such as surveying, remote sensing, radar, communication, and industrial processing [1], [2], [3]. The effectiveness of microwave applications is closely tied to the electromagnetic properties of materials, particularly their complex permittivity and permeability. In industrial microwave energy applications, the complex permittivity plays a pivotal role in determining system performance. As fundamental parameters characterizing a material's interaction with electromagnetic waves, complex dielectric constants are crucial for evaluating the energy storage and loss capabilities of materials

in the presence of an electric field. Consequently, accurate measurement of complex permittivity is significant in both material science and microwave engineering.

Compared with traditional characterization techniques, microwave-based sensing methods have gained increasing prominence for permittivity measurement due to their inherent advantages, including high sensitivity, low cost, and structural simplicity [4]. These methods generally rely on extracting complex dielectric constants from measured scattering parameters (S -parameters). Owing to their strong penetration capability and high detection sensitivity, microwave sensors have been successfully used in various application domains, including biomedical sensing and liquid analysis [5], [6], [7], [8], [9], [10].

In recent years, numerous techniques have been proposed for the characterization of dielectric constants, including the free-space method, the transmission line method, and the resonance-based method. Among these, microwave planar resonators leveraging resonant phenomena have garnered significant attention due to their compact size, low cost, and ease of integration into planar circuits. Common resonator structures adopted in planar microwave sensors include the split-ring resonator (SRR) [11], [12], complementary split-ring resonator (CSRR) [13], interdigital capacitor [14], balanced

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In summary, recent advances in microwave planar resonator design have primarily focused on enhancing sensitivity and improving the accuracy of complex permittivity extraction through *S*-parameter inversion. In this work, a novel microwave planar sensor based on a hexagonal CSRR (H-CSRR) is proposed. Compared with conventional circular geometries, the hexagonal structure provides a stronger and more localized electric field, thereby improving the sensor's sensitivity. To further enhance the field confinement, an IDE structure is embedded between the two hexagonal rings. In addition, a metallic via is introduced between the resonator and the microstrip line to form an equivalent inductive element, which serves to strengthen the coupling effect and lower the overall resonant frequency. This configuration not only enhances the interaction between the resonator and the material under test but also mitigates the influence of the real part of the complex permittivity on the inversion of its imaginary part, thereby improving the overall accuracy of dielectric characterization. Unlike prior single-innovation approaches, this work synergistically combines geometric field enhancement (H-CSRR), capacitive focusing (IDE), and coupling enhancement (via inductance) to simultaneously address sensitivity and accuracy limitations—establishing a new design paradigm for permittivity sensors.

A. Basic Principle of Sensors

$$\frac{\Delta f_r}{f_r} = \frac{\int_v (\Delta \varepsilon \vec{E}_1 \cdot \vec{E}_0 + \Delta \mu \vec{H}_1 \cdot \vec{H}_0) dv}{\int_v (\varepsilon_0 |\vec{E}_0|^2 + \mu_0 |\vec{H}_0|^2) dv} \quad (1)$$

unloaded conditions, respectively, $\vec{E}_1$ and $\vec{H}_1$ represent those under loaded conditions at resonance. The parameters ϵ_0 and μ_0 indicate the permittivity and permeability of free space, respectively. $\Delta\epsilon$ and $\Delta\mu$ denote the changes in permittivity and magnetic permeability, respectively, while v represents the volume of the perturbation.

$$\frac{\Delta f_r}{f_r} = \frac{\int_v (\Delta \varepsilon \vec{E}_1 \cdot \vec{E}_0) dv}{\int_v (\varepsilon_0 |\vec{E}_0|^2) dv}. \quad (2)$$

According to (2), when the electric field changes due to the introduction of the measured material, the greater the change, the greater the frequency shift. Therefore, the electric field strength during resonance should be maximized as far as possible, and the measured material should be located at the maximum field strength.

The proposed microwave planar sensor is a low-cost F4B substrate with a thickness of 1 mm and a dielectric constant of 2.65. Fig. 1(a) shows the design of the sensor structure. The resonator structure is etched on the bottom metal plane, and the top of the substrate is a 50- Ω microstrip line. The resonator is excited by capacitive coupling with the microstrip line, and equivalent inductance coupling is introduced by the metal through the hole. The geometric parameters of the sensor structure are shown in Table I. The comparison of the electric field distribution between the sensor and the traditional CSRR sensor is shown in Fig. 2. Compared with the traditional CSRR sensor, the sensor proposed in this article has a larger electric field intensity, which is mainly concentrated between two hexagonal rings. Therefore, this area is taken as the placement area of the measurement sample, as shown in Fig. 1(b).

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TABLE I
GEOMETRICAL PARAMETERS OF THE SENSOR UNIT: mm

W	R_1	R_2	D_1	D_2
2.65	5	3.2	1	1
W_1	W_2	W_{IDE}	l_{IDE}	d
0.3	0.7	0.2	0.7	0.6

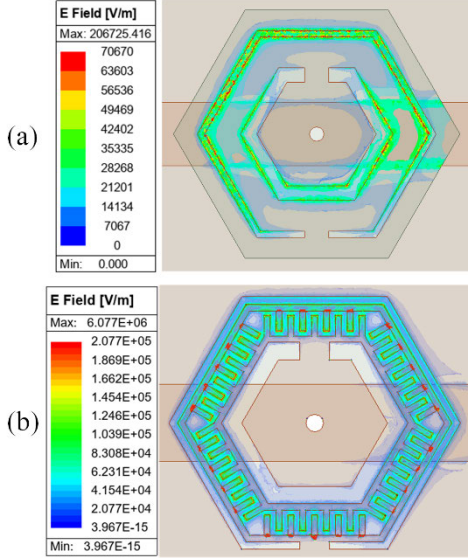


Fig. 2. Simulation of the electric field at f_0 . (a) Conventional sensor. (b) Sensor loaded with the IDE.

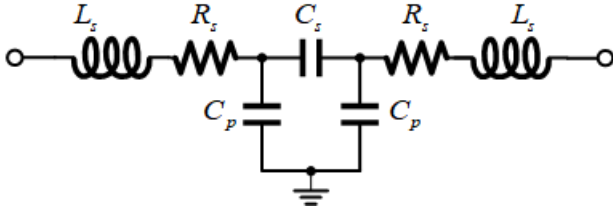


Fig. 3. IDE structure equivalent circuit.

diagram of the IDE structure is shown in Fig. 3, where L_s and R_s are equivalent inductance and equivalent resistance, C_s and C_p represent series capacitance and parallel capacitance of the IDE structure, respectively.

The equivalent capacitance of the IDE structure can be expressed by (3), where ϵ_0 is the dielectric constant of the dielectric substrate, K and $K(k)$ are the first type of elliptic integral, W_{IDE} , W_d , and L_{IDE} represent the width, gap, and length of the interfinger structure, respectively, and N represents the total number of interfingers of the IDE structure

$$C_{IDE} = \left[\epsilon_0 \left(\frac{\epsilon_{Sub} + 1}{2} \right) \times \frac{K \sqrt{K^2 + 1}}{K(k)} + \epsilon_0 \frac{W_{IDE}}{W_d} \right] \times (N - 1) L_{IDE}. \quad (3)$$

The simulation results of the hexagonal CSRR sensor loaded with the IDE structure are shown in Fig. 4.

It can be seen that under the condition of no loss, the sensor's frequency decreases from 3.47 to 2.48 GHz with the

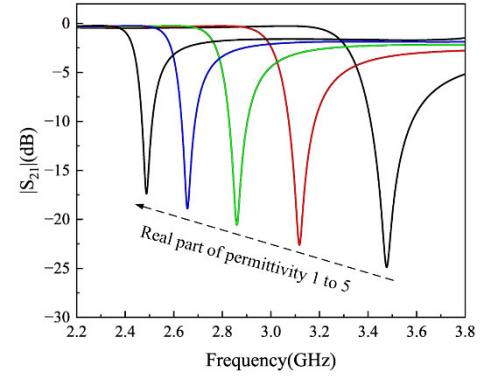


Fig. 4. Simulation results of the H-CSRR sensor loaded with the IDE.

change in the real part of the relative dielectric constant, and $|S_{21}|$ decreases by 7.4 dB. However, in previous studies, the corresponding relationship between the real part of the relative dielectric constant and the resonant frequency, and the imaginary part and $|S_{21}|$ has often been established, respectively. However, according to the simulation results, $|S_{21}|$ will be affected by the real part, as shown in the following equation:

$$|S_{21}| = F(\epsilon_r', \epsilon_r''). \quad (4)$$

Therefore, establishing the relationship between the imaginary and real parts is not rigorous, and the established relationship is less generalized.

In this article, by introducing a metal through-hole with equivalent inductance between the resonator and the microstrip line, not only is the overall resonant frequency reduced but also the influence of the real part of the relative dielectric constant is reduced. The simulation results of the new structure are shown in Fig. 5. The resonant frequency of the sensor decreases from 2.79 to 1.96 GHz with the change in the real part of the relative dielectric constant, and the change in $|S_{21}|$ is only 0.7 dB. It can be approximated that ϵ_r' has nothing to do with $|S_{21}|$. It facilitates the establishment of a more rigorous and generalized correspondence between ϵ_r' and $|S_{21}|$.

Compared with the sensor without a metal through-hole, the sensor has higher sensitivity, a more accurate inversion effect for the imaginary part, and stronger universality.

C. Equivalent Circuit Model

The proposed sensor equivalent circuit model is shown in Fig. 6, where Z_0 is the characteristic impedance of the microstrip line and L is the inductance of the microstrip line. C_0 and L_0 represent the coupling capacitance and coupling inductance between the microstrip line and the resonant structure, respectively. C_c , L_c , and R_c represent the equivalent capacitance, inductance, and resistance of the CSRR structure, respectively.

The formula for calculating the resonant frequency of the sensor is shown in the following equation:

$$f = \frac{1}{2\pi \sqrt{\frac{L_c L_0}{L_c + L_0} (C_0 + C_c + 2C_p)}}. \quad (5)$$

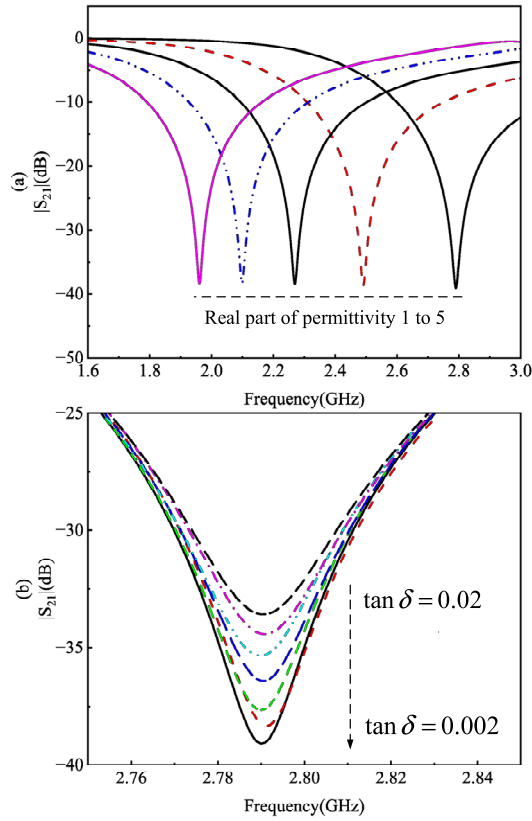


Fig. 5. Simulation of the sensor with a metal through-hole. (a) Lossless samples. (b) Lossy samples.

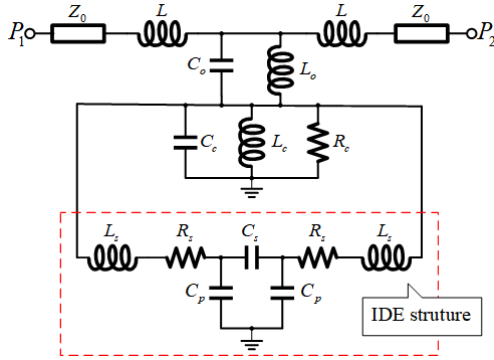


Fig. 6. Equivalent circuit model.

To verify the accuracy of circuit topology, ADS software was used to simulate the equivalent circuit lumped element parameters, as shown in Table II.

The simulation results of the circuit model were compared with the electromagnetic simulation results, as shown in Fig. 7.

III. EXPERIMENT AND RESULT ANALYSIS

A. Sensitivity Analysis

The sensor is used for dielectric constant measurement, so high sensitivity is required. Sensitivity is defined as

$$S = \frac{\Delta f}{\Delta \varepsilon} \quad (6)$$

where $\Delta \varepsilon$ is the change in the relative dielectric constant. However, sensors operating at high frequencies tend to exhibit

TABLE II
EQUIVALENT CIRCUIT LUMPED ELEMENT PARAMETERS

L (nH)	C_o (pF)	L_o (nH)	C_c (pF)	L_c (nH)
1.83	1.48	39.40	0.85	1.38
R_c (k Ω)	L_s (nH)	R_s (Ω)	C_s (pF)	C_p (pF)
4.64	0.95	3.00	8.43	0.06

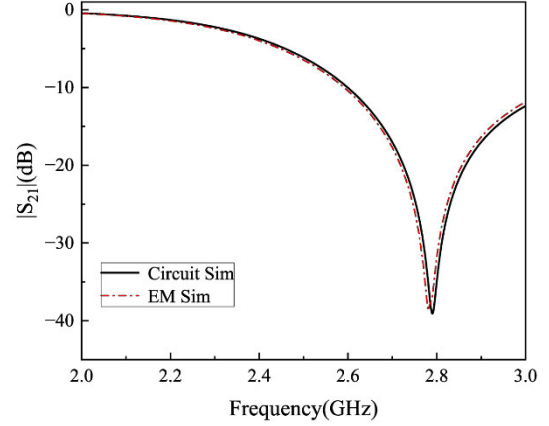


Fig. 7. Comparison between the circuit simulation model and electromagnetic simulation model.

higher Δf , so we standardize the sensitivity, as shown in the following equation:

$$S = \frac{\Delta f}{f_0 \Delta \varepsilon}. \quad (7)$$

B. Experiment and Result Analysis

To verify the sensor analyzed and proposed in this article, we manufactured the sensor. The welded SMA interface is connected to the VNA, and the frequency change with $|S_{21}|$ is observed on the VNA. Fig. 8 shows the processed sensor and the measurement environment.

During measurements, all the MUTs were standardized to dimensions of $15 \times 15 \times 3$ mm. A plastic clamp was used to secure both the MUT and the sensor, minimizing air gaps and ensuring complete contact between them. Five common composite materials with distinct complex dielectric constant values were tested, with the measurement results in Fig. 9.

According to Fig. 5, the nonlinear least-square method is used to fit the curve, and the relationship is as follows:

$$\varepsilon'_r = 2.62 \times \Delta f^3 - 0.47 \times \Delta f^2 + 3.48 \times \Delta f + 0.93 \quad (8)$$

$$\varepsilon''_r = 0.000022 \times (\Delta |S_{21}|)^3 + 0.00011 \times (\Delta |S_{21}|)^2 + 0.0046 \times \Delta |S_{21}| - 0.0014 \quad (9)$$

where Δf is the resonant frequency shift and $\Delta |S_{21}|$ is the difference between $|S_{21}|$ with loaded and unloaded samples.

TABLE III
MEASURED VALUE OF THE MATERIAL TO BE TESTED IS COMPARED WITH OTHER RESEARCH RESULTS

MUT	Reference value of the specimen			Measured data		Error		Sensitivity $S\%$
	ϵ_r'	$\tan \delta$	ϵ_r''	ϵ_r'	ϵ_r''	$\epsilon_r' \%$	$\epsilon_r'' \%$	
PTFE	2.1	0.00028	0.00058	2.1151	0.000577	0.72	0.52	10.55
PP	2.28	0.00008	0.00018	2.2780	0.000178	0.088	1.11	10.16
ABS	2.4	0.0125	0.03	2.3881	0.0317	0.50	5.67	9.95
PET	2.87	0.00836	0.024	2.8024	0.0249	2.4	3.75	9.16
FR4	4.4	0.02	0.088	4.5166	0.0916	2.7	4.09	8.01

TABLE IV
COMPARISON OF THE PROPOSED SENSORS TO RECENT RESEARCH WORK

Ref	Resonant type	f_0 (GHz)	S_{av} (%)	ϵ_r' Meas. Error	ϵ_r'' Meas. Error
[22]	MCSRR	2.88	4.49	7.48%	N/A
[23]	CSRR	4.05	4.76	9.09%	N/A
[23]	CSRR	2.45	4.12	0.542%	N/A
[16]	P-IDC	2.51	3.98	1%	3%
[24]	EBG	2.4	N/A	0.17%	3.33%
[25]	CMC-SRR	17	8.05	3.1%	N/A
[18]	CSRR	2.46	0.448	4.6%	5.1%
[26]	CSRR-IDE	1.8	2.64	4%	12.4%
[27]	CSRR	4.24	4.05	3.34%	N/A
[28]	M-shaped slot	2.459	2.9	N/A	N/A
[29]	Ring-resonator	2.72	0.726	2.4%	5.8%
This work	CSRR-IDE	2.82	9.566	1.28	3.028%

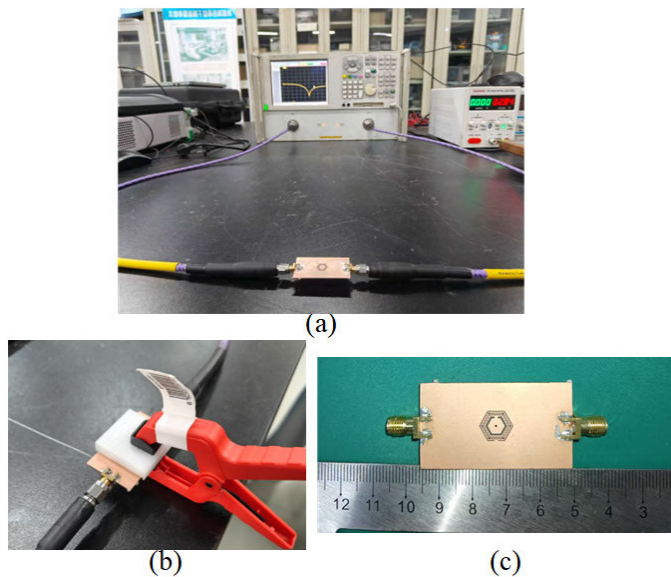


Fig. 8. Sensors and the measurement system. (a) Measurement system. (b) Sensor loaded with sample. (c) Fabricated sensor.

To validate the stability and repeatability of the proposed sensor, five independent experiments were conducted.

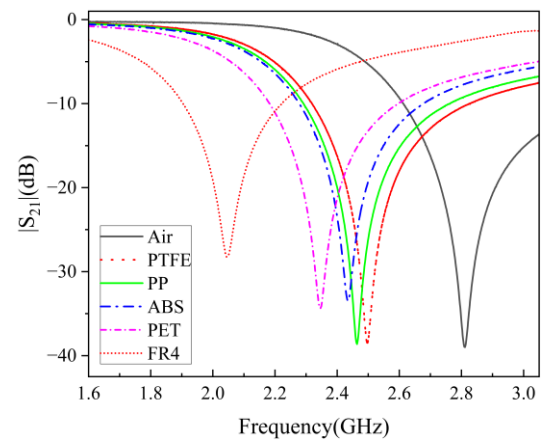


Fig. 9. Measurement results of the samples.

Statistical analysis was performed on complex permittivity measurements, with the results presented in Fig. 10 and the corresponding errors shown in Fig. 11. The error bars for the imaginary part of complex permittivity were scaled by a factor of 2 to enhance clarity in the plot. The observed variations are minimal, suggesting good stability of the proposed sensor.

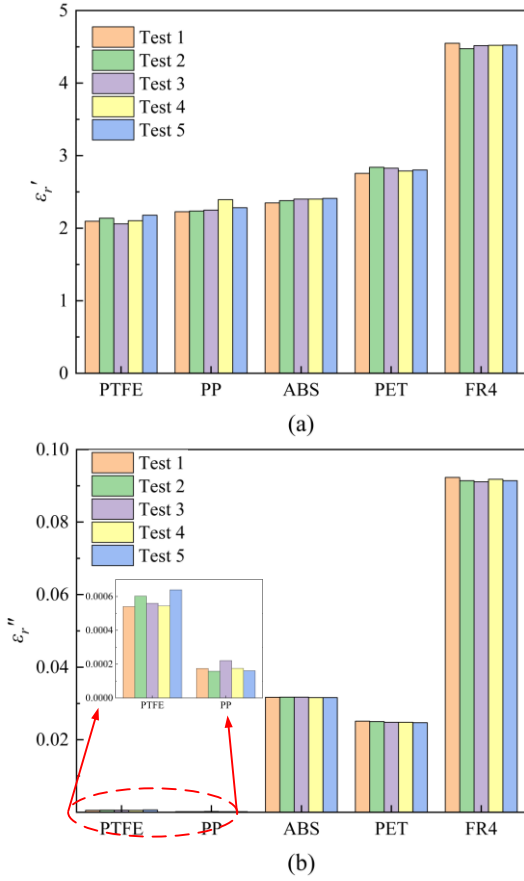


Fig. 10. Measurements of five different samples. (a) Real part and (b) imaginary part of the dielectric constant.

The real and imaginary parts of the complex permittivity are derived by substituting the statistically averaged Δf and $\Delta|S_{21}|$ values from five experimental trials into (8) and (9).

For each MUT measurement, both sensitivity and measurement error were calculated. The error was determined using the following formula:

$$\text{err}_{\epsilon'_r} = \frac{|\epsilon'_{\text{test}} - \epsilon'_{\text{act}}|}{\epsilon'_{\text{act}}} \times 100\% \quad (10)$$

$$\text{err}_{\epsilon''_r} = \frac{|\epsilon''_{\text{test}} - \epsilon''_{\text{act}}|}{\epsilon''_{\text{act}}} \times 100\% \quad (11)$$

where ϵ'_{test} and ϵ''_{test} represent the real and imaginary parts of the complex permittivity of MUT calculated according to the test results, respectively, and ϵ'_{act} and ϵ''_{act} represent the real and imaginary parts of the actual complex permittivity of MUT, respectively [19], [20], [21]. The calculation results are shown in Table III.

Table IV shows comparison of the proposed sensor with other recently proposed sensors of similar structures [16], [18], [22], [23], [24], [25], [26], [27], [28], [29]. The proposed sensor achieves a breakthrough in sensitivity while enabling simultaneous measurement of both the real and imaginary parts of complex permittivity, maintaining low measurement errors. The average sensitivity is 9.6%, and the average error of the real and imaginary parts is 1.3% and 3.0%, respectively. This sensor shows strong competitiveness.

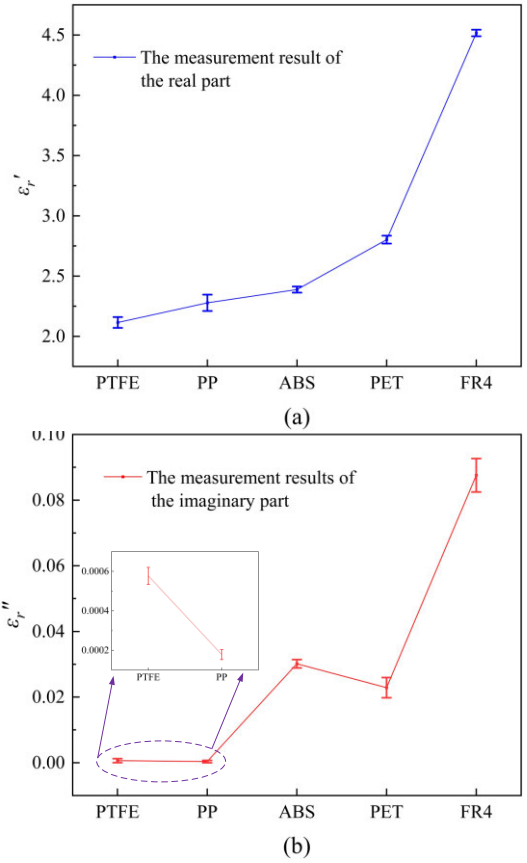


Fig. 11. Error bars of the five experimental tests. (a) Real part and (b) imaginary part of the dielectric constant.

IV. CONCLUSION

This article proposes a low-cost microwave planar sensor based on the CSRR-IDE structure to characterize the complex dielectric constant of solid materials. The designed resonator part comprises a CSRR-loaded IDE structure to generate a high electric field intensity in its sensing area. The resonator is excited through the capacitive coupling between the microstrip line and the inductance coupling introduced by the metal through-hole. The excitation method can not only enhance the coupling and reduce the frequency but also reduce the influence of the real part of the complex dielectric constant on $|S_{21}|$. The experimental results show not only excellent sensitivity (9.6%) but also small errors for the measurement of the real part (1.3%) and the imaginary part (3.0%) of the dielectric constant. The proposed sensor provides a more concise and accurate inversion equation in numerical modeling, shows small errors in measurements, presents excellent competitiveness, and has good application prospects in industry, medicine, and other fields.

REFERENCES

- [1] C. Liu and X. Wu, "Research progress on industrial application of microwave energy," *Inf. Electron. Eng.*, vol. 10, no. 4, pp. 451–455, 2012.
- [2] R. Tang, C. Liu, C. Liao, H. Wen, B. Wu, and C. Wang, "Study on S-band frequency conversion microwave square cavity dry PET slice," *Packag. Eng.*, vol. 45, no. 5, pp. 1–7, 2024, doi: 10.19554/j.cnki.1001-3563.2024.05.001.
- [3] C. Liao, Z. He, R. Tang, W. Zhang, C. Liu, and C. Wang, "The research on microwave drying characteristics of polyethylene terephthalate materials based on frequency and power tuning technology," *Processes*, vol. 12, no. 7, p. 1488, Jul. 2024.

- [4] C. Liu, C. Liao, Y. Peng, W. Zhang, B. Wu, and P. Yang, "Microwave sensors and their applications in permittivity measurement," *Sensors*, vol. 24, no. 23, p. 7696, Dec. 2024.
- [5] Z. Long and C. Liu, "A dual-frequency substrate integrated waveguide dielectric constant measurement system is presented," *Appl. Technol.*, vol. 46, no. 3, pp. 21–24, 2019.
- [6] C. Liu and F. Tong, "An SIW resonator sensor for liquid permittivity measurements at C band," *IEEE Microw. Wireless Compon. Lett.*, vol. 25, no. 11, pp. 751–753, Nov. 2015, doi: [10.1109/LMWC.2015.2479851](https://doi.org/10.1109/LMWC.2015.2479851).
- [7] C. Liao, L. Jiang, Q. Qu, Q. Chen, and C. Liu, "The invention relates to a complex dielectric constant measuring sensor with a substrate integrated waveguide terminal loading structure," *Appl. Technol.*, vol. 48, no. 6, pp. 13–17, 2021.
- [8] M. A. Ahmad, H. A. Qamar, K. Mohammed, S. Meetani, and R. A. Ketbi, "Blood flow detection using piezoelectric sensors," *IEEE Sensors J.*, vol. 25, no. 3, pp. 4316–4323, Feb. 2025, doi: [10.1109/JSEN.2024.3517166](https://doi.org/10.1109/JSEN.2024.3517166).
- [9] K. Ma, C. Liu, and Q. Chen, "The utility model relates to a dielectric constant measuring device with a dielectric integrated suspension line structure," *Appl. Technol.*, vol. 48, no. 4, pp. 29–33, 2021.
- [10] H. Chen, Q. Chen, C. Liu, K. Huang, and Z. Long, "Wideband dielectric coefficient measuring device based on SIW," *J. Chem. Eng.*, vol. 70, no. 1, pp. 182–185, 2019.
- [11] M. Bazgir and A. Sheikhi, "High Q-factor compact permittivity sensor based on coupled SRR-ELC metamaterial element and metasurfaces shield," *IEEE Sensors J.*, vol. 24, no. 4, pp. 4424–4431, Feb. 2024, doi: [10.1109/JSEN.2023.3345477](https://doi.org/10.1109/JSEN.2023.3345477).
- [12] D. Prakash and N. Gupta, "Microwave grooved SRR sensor for detecting low concentration ethanol-blended petrol," *IEEE Sensors J.*, vol. 23, no. 14, pp. 15544–15551, Jul. 2023, doi: [10.1109/JSEN.2023.3279851](https://doi.org/10.1109/JSEN.2023.3279851).
- [13] H.-Y. Gan et al., "Differential microwave microfluidic sensor based on microstrip complementary split-ring resonator (MCSRR) structure," *IEEE Sensors J.*, vol. 20, no. 11, pp. 5876–5884, Jun. 2020, doi: [10.1109/JSEN.2020.2973196](https://doi.org/10.1109/JSEN.2020.2973196).
- [14] J. Zhang, G. Liu, S. Jiang, Y. Gong, and R. Zhang, "A microwave sensor based on grounded coplanar waveguide for solid material measurement," *IEEE Sensors J.*, vol. 25, no. 3, pp. 4192–4200, Feb. 2025, doi: [10.1109/JSEN.2024.3516757](https://doi.org/10.1109/JSEN.2024.3516757).
- [15] Y. Kato and M. Horibe, "Broadband permittivity measurements up to 170-GHz using balanced-type circular-disk resonator excited by 0.8-mm coaxial line," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 6, pp. 1796–1805, Jun. 2019, doi: [10.1109/TIM.2018.2886864](https://doi.org/10.1109/TIM.2018.2886864).
- [16] C. Wang et al., "High-accuracy complex permittivity characterization of solid materials using parallel interdigital capacitor-based planar microwave sensor," *IEEE Sensors J.*, vol. 21, no. 5, pp. 6083–6093, Mar. 2021, doi: [10.1109/JSEN.2020.3041014](https://doi.org/10.1109/JSEN.2020.3041014).
- [17] S. Jiang, G. Liu, M. Wang, Y. Wu, and J. Zhou, "Design of high-sensitivity microfluidic sensor based on CSRR with interdigital structure," *IEEE Sensors J.*, vol. 23, no. 16, pp. 17901–17909, Aug. 2023, doi: [10.1109/JSEN.2023.3294243](https://doi.org/10.1109/JSEN.2023.3294243).
- [18] X. Han et al., "Highly integrated improved hexagonal CSRR-based fluid sensor for complex dielectric parameter detection," *IEEE Sensors J.*, vol. 24, no. 13, pp. 20559–20570, Jul. 2024, doi: [10.1109/JSEN.2024.3399454](https://doi.org/10.1109/JSEN.2024.3399454).
- [19] J. Krupka, "Measurements of the complex permittivity of low loss polymers at frequency range from 5 GHz to 50 GHz," *IEEE Microw. Wireless Compon. Lett.*, vol. 26, no. 6, pp. 464–466, Jun. 2016, doi: [10.1109/LMWC.2016.2562640](https://doi.org/10.1109/LMWC.2016.2562640).
- [20] J. Zechmeister and J. Lacik, "Complex relative permittivity measurement of selected 3D-printed materials up to 10 GHz," in *Proc. Conf. Microw. Techn. (COMITE)*, Apr. 2019, pp. 1–4, doi: [10.1109/COMITE.2019.8733590](https://doi.org/10.1109/COMITE.2019.8733590).
- [21] B. Riddle, J. Baker-Jarvis, and J. Krupka, "Complex permittivity measurements of common plastics over variable temperatures," *IEEE Trans. Microw. Theory Techn.*, vol. 51, no. 3, pp. 727–733, Mar. 2003, doi: [10.1109/TMTT.2003.808730](https://doi.org/10.1109/TMTT.2003.808730).
- [22] A. Hanif et al., "Highly sensitive dual-notch compact complementary meta resonator for low permittivity," *IEEE Sensors J.*, vol. 24, no. 20, pp. 32313–32320, Oct. 2024, doi: [10.1109/JSEN.2024.3450930](https://doi.org/10.1109/JSEN.2024.3450930).
- [23] X. Han et al., "CSRR metamaterial microwave sensor for measuring dielectric constants of solids and liquids," *IEEE Sensors J.*, vol. 24, no. 9, pp. 14167–14176, May 2024, doi: [10.1109/JSEN.2024.3373755](https://doi.org/10.1109/JSEN.2024.3373755).
- [24] S. Subbaraj, V. S. Ramalingam, M. Kanagasabai, E. F. Sundarsingh, Y. P. Selvam, and S. Kingsley, "Electromagnetic nondestructive material characterization of dielectrics using EBG based planar transmission line sensor," *IEEE Sensors J.*, vol. 16, no. 19, pp. 7081–7087, Oct. 2016, doi: [10.1109/JSEN.2016.2591320](https://doi.org/10.1109/JSEN.2016.2591320).
- [25] T. Haq and S. Koziel, "Novel complementary multiple concentric split ring resonator for reliable characterization of dielectric substrates with high sensitivity," *IEEE Sensors J.*, vol. 24, no. 10, pp. 16233–16241, May 2024, doi: [10.1109/JSEN.2024.3386248](https://doi.org/10.1109/JSEN.2024.3386248).
- [26] L. Ali et al., "Simultaneous measurement of thickness and permittivity using microwave resonator-based planar sensor," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 31, no. 10, p. 22794, Oct. 2021.
- [27] N. C. Pradhan and S. Koziel, "Fully 3D-printed microwave sensor for permittivity measurement," *IEEE Sensors J.*, vol. 25, no. 12, pp. 21546–21551, Jun. 2025, doi: [10.1109/JSEN.2025.3567295](https://doi.org/10.1109/JSEN.2025.3567295).
- [28] A. Rasoulzadeh, C. Ghobadi, J. Nourinia, and P. Mohammadi, "A highly sensitive and compact size microstrip sensor for characterization of solid and liquid material with broad range permittivity," *IEEE Access*, vol. 12, pp. 87864–87872, 2024, doi: [10.1109/ACCESS.2024.3416392](https://doi.org/10.1109/ACCESS.2024.3416392).
- [29] F. Jafari and M. R. Malekshahi, "A low-cost microwave microfluidic sensor based on planar ring resonator," *IEEE Sensors J.*, vol. 23, no. 18, pp. 21070–21077, Sep. 2023, doi: [10.1109/JSEN.2023.3301813](https://doi.org/10.1109/JSEN.2023.3301813).



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